

## Problem Statement

The Federal Trade Commission (FTC) is a government agency that is responsible for "*protecting consumers and competition by preventing anticompetitive, deceptive, and unfair business practices through law enforcement, advocacy, and education without unduly burdening legitimate business activity.*"<sup>1</sup> It conducts investigative cases against companies that engage in unlawful practices that harm consumers in any way. These cases are logged in [FTC site](https://www.ftc.gov/about-ftc) and are available for public review.

You need to build a tool that can help analyze these FTC cases. To start with, Table 1 shows a sample of cases from the dataset. The data is in a tab-separated file named FTC-cases-TSV.txt. Important things to note in data:

1. Each case has a date, a title, a case type, and a case number. However, case type and case number may be missing for some cases.
2. The date is in yyyy-mm-dd format
3. Case type is given at the end of the Title within parentheses. The current data has Federal and Administrative as two case types, or the case type could be missing.
4. Case number is a string of some characters. It could be something like "1823184", or "192 3133", or "182 3106; 182 3107" or "092 3184; 182 3109; C-4365"

Table 1: FTC Case Data

| Case date  | Title (case type)                                                  | Case number                |
|------------|--------------------------------------------------------------------|----------------------------|
| 2021-07-01 | Kuuuuub, Inc., et al., U.S. v. (Recolor Oy) (Federal)              | 1823184                    |
| 2021-06-22 | Flo Health, Inc. (Administrative)                                  | 192 3133                   |
| 2021-05-07 | Everalbum, Inc., In the Matter of (Administrative)                 | 192 3172                   |
| 2021-05-03 | Support King, LLC (SpyFone.com), In the Matter of (Administrative) | 192 3003                   |
| 2021-04-29 | Vivint Smart Home, Inc. (Federal)                                  | 192 3060                   |
| 2020-05-20 | Tapplock, Inc., In the Matter of (Administrative)                  | 192 3011                   |
| 2020-04-28 | Facebook, Inc., In the Matter of (Administrative)                  | 092 3184; 182 3109; C-4365 |

The first version of our tool needs to do the following:

1. Search for cases reported against a company
2. Search for cases reported in a year
3. Search for a case with a case number
4. Print case type summary
5. Print year-wise summary

Please refer to sample scenarios at the end of this document.

## Solution Design

The solution design has four classes.

<sup>1</sup> <https://www.ftc.gov/about-ftc>

### CyberCop

- This class will handle user interactions and outputs.

### CCModel:

- `loadCases()` reads data from the input file to create Case objects and stores the cases in an array named *cases*.

### SearchEngine:

- `searchTitle()`: takes `searchString` and `cases` array. Returns an array of Cases whose title contains the `searchString`. The search is case-insensitive
- `searchYear()`: takes `year` as a String and `cases` array. Returns an array of Cases that the case falls in the year
- `searchCaseNumber`: takes `caseNumber` and `cases` array and return an array of Cases that contain the `caseNumber` in their `caseNumber` property.

### Case:

- The case should have instance members of *caseDate*, *caseTitle*, *caseType* and *caseNumber* which are all Strings

**Design Constraint:** Use of Collection classes (ArrayLists, LinkedLists, Sets, Maps, etc.) is not allowed in HW1.

### Instructions:

- Complete your code.
- You will test your program in two ways: Console output (refer sample outputs at the end of this document) and `TestCyberCop.java`.

See Homework 1 page in Canvas for the grading criteria.

**NO LATE SUBMISSIONS PLEASE!** If you don't submit on time, you lose all the points. Please avoid last minute submission as Canvas may decide to quit on you! **No late submission will be accepted.**

**Sample outputs****Scenario 1: Menu option 1 - A company has multiple cases**

```

*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
1
-----
Enter search string
google
-----
3 case(s) found for google
-----
#. Last update Case Title                                Case Type      Case/File Number
-----
1. 2019-09-04  Google LLC and YouTube, LLC              Federal        172 3083
2. 2012-11-20  Google Inc.                                Federal        102 3136
3. 2011-10-24  Google, Inc., In the Matter of                     Administrative
-----

```

**Scenario 2: Menu option 1 - A company with no reported case**

```

*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
1
-----
Enter search string
apple
-----
Sorry, no search results found for apple
-----

```

## Scenario 3: Menu option 2 - Several cases reported in a year

\*\*\* Welcome to CyberCop! \*\*\*

1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit

2

-----  
Enter search year as YYYY

2021

5 case(s) found for 2021

| #. Last update | Case Title                                        | Case Type      | Case/File Number |
|----------------|---------------------------------------------------|----------------|------------------|
| 1. 2021-07-01  | Kuuuub, Inc., et al., U.S. v. (Recolor Oy)        | Federal        | 1823184          |
| 2. 2021-06-22  | Flo Health, Inc.                                  | Administrative | 192 3133         |
| 3. 2021-05-07  | Everalbum, Inc., In the Matter of                 | Administrative | 192 3172         |
| 4. 2021-05-03  | Support King, LLC (SpyFone.com), In the Matter of | Administrative | 192 3003         |
| 5. 2021-04-29  | Vivint Smart Home, Inc.                           | Federal        | 192 3060         |

## Scenario 4: Menu option 2 - No cases in a year

```

*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
2
-----
Enter search year as YYYY
-----
1994
Sorry, no search results found for 1994
-----

```

## Scenario 5: Menu option 3 - Case number found

```

*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
3
-----
Enter case number
-----
092 3184
1 case(s) found for 092 3184
-----
#.# Last update Case Title
-----
1. 2020-4-28 Facebook, Inc., In the Matter of
-----

```

| Case Type      | Case/File Number           |
|----------------|----------------------------|
| Administrative | 092 3184; 182 3109; C-4365 |

Scenario 6: Menu option 3 - Case number not found

```
*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
3
-----
Enter case number
-----
12345
Sorry, no search results found for 12345
-----
```

Scenario 7: Menu option 4

```
*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
4
-----
*** Case Type Summary Report ***
No. of Administrative cases: 161
No. of Federal cases: 115
No. of unknown case types: 6
-----
```

## Scenario 8: Menu option 5

```
*** Welcome to CyberCop! ***
1. Search cases for a company
2. Search cases in a year
3. Search case number
4. Print Case-type Summary
5. Print Year-wise Summary
6. Exit
```

```
5
```

```
-----
*** Year-wise Summary Report ***
*** Number of FTC cases per year ***
```

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| 2021: 5  | 2020: 25 | 2019: 20 | 2018: 17 | 2017: 9  |
| 2016: 9  | 2015: 21 | 2014: 30 | 2013: 17 | 2012: 12 |
| 2011: 16 | 2010: 16 | 2009: 11 | 2008: 12 | 2007: 5  |
| 2006: 7  | 2005: 8  | 2004: 10 | 2003: 7  | 2002: 8  |
| 2001: 4  | 2000: 7  | 1999: 3  | 1998: 1  | 1997: 2  |

```
-----
```